DATA ANALYST PROJECT · POSTGRESQL · POWER BI

India Oil, Rupee & Inflation Case Study

A PostgreSQL database and Power BI dashboard that test how crude oil prices relate to India's rupee, inflation, interest rates and stock market, using seven real datasets from 1973 to 2026 and ten SQL analysis questions.

DATABASE ENGINE
PostgreSQL
pgAdmin 4

SCHEMA
india_oil_rupee_inflation
7 tables · 28,075 rows

ANALYSIS
10 SQL questions
4 analysis methods

DASHBOARD
Power BI
6 KPI cards · 6 charts

CONCEPTS COVERED
Joins · CTEs · Window functions
LAG, LEAD, rolling CORR(), event studies

DATA WINDOW
2007 – 2026
Series start 1973 (USD/INR)

WHAT THIS DOCUMENT CONTAINS

The schema and data sources, a data-quality review, the starter exploration queries, all ten analysis questions with their full SQL, result tables and plain-language findings, the Power BI dashboard, and a closing section with the key findings, limitations and a run guide. Every number in the result tables was produced from the project database. The project looks for patterns and relationships; it does not claim that one variable causes another.

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Sixteen parts: the project set-up, ten analysis questions, the dashboard and the closing summary.

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SECTION 01

Project Overview

A single PostgreSQL database that brings India's oil, currency, inflation, interest-rate, fuel-price and stock-market data together, then answers ten business questions about how they move relative to each other.

The `india_oil_rupee_inflation` database holds seven public data series, from daily Brent crude prices to monthly inflation indices and event-based RBI rate changes. A set of ten analysis questions, each with its SQL, result table and a plain-language finding, tests how the series relate to one another. A one-page Power BI dashboard then summarises the project for a non-technical reader.

7	28,075	10
BASE TABLES	ROWS LOADED	ANALYSIS QUESTIONS
4	**6**	**6**
ANALYSIS METHODS	DASHBOARD KPI CARDS	DASHBOARD CHARTS

Hypothesis chain tested

OIL	RUPEE	INFLATION	RBI POLICY	STOCK MARKET
Brent crude	**USD / INR**	**CPI and WPI**	**Repo rate**	**NIFTY 50, Sensex**
`crude_oil_data`	`currency_data`	`inflation_data` `wholesale_price_data`	`monetary_policy_data`	`stock_market_data`

Patterns only, not proof of cause. The project identifies relationships and patterns and does not assume that one variable directly causes another.

Goal

- Analyse how changes in crude oil prices, the Indian rupee, inflation and RBI monetary policy are related to each other and to the Indian stock market.
- Test each link with one of four methods: trend analysis, correlation analysis, event study and regime comparison.
- State the thresholds, sample sizes and data gaps next to every result, so a reader can judge how far each finding can be trusted.
- Present the outcome in a one-page Power BI dashboard built on the same database tables.

Project at a glance

Item	Detail
Topic	How crude oil prices relate to India's rupee, inflation, interest rates and stock market
Data window used for most questions	2007-01 to 2026-09 (NIFTY 50 starts in 2007); Brent from 1987 and USD/INR from 1973 are used where no stock data is needed
Database	PostgreSQL, schema `india_oil_rupee_inflation` , 7 tables, 28,075 rows
Questions	10 analysis questions, each with SQL, a result table and a plain-language finding
Dashboard	One-page Power BI dashboard: 6 KPI cards and 6 charts

Analysis types used

Method	What it answers	Questions
Trend analysis	How did oil, the rupee and the market change year by year?	Q1, Q9
Correlation analysis	Do two series move together, and with what delay?	Q2, Q5
Event study	What happens after an oil spike, a rupee shock or an RBI rate decision?	Q3, Q4, Q7
Regime comparison	How do things differ in high-oil vs normal-oil periods, or hiking vs cutting months?	Q6, Q8, Q10

SQL techniques demonstrated

Area	Applied in this project
Aggregation	`AVG` , `COUNT` , `SUM` , `STDDEV` , `GROUP BY` , `DATE_TRUNC` , `EXTRACT`
Common table expressions	Multi-block `WITH` queries chaining 5 to 9 steps
Window functions	`LAG` and `LEAD` (1 to 6 steps), rolling `CORR()` over 36 months
Conditional logic	`CASE` labels for sharp years, oil spikes, RBI stance and oil-price buckets
Joins	INNER and LEFT joins across daily, monthly and event-based tables
Statistics in SQL	`CORR()` , `STDDEV()` , percentage change, ranking and decomposition
Data quality	NULL handling, gap markers, duplicate-date checks, overlap-window analysis

SECTION 02

Data Sources and Schema

Seven tables arranged by how often their data changes: daily market series, monthly price indices, and two event-based tables that only record a row when something changed.

Table inventory

Table	Rows	Date range	Frequency	Source
`crude_oil_data`	9,097	1987-05-20 to 2026-09-29	Daily	US EIA – Europe Brent Spot Price FOB
`currency_data`	13,473	1973-01-02 to 2026-10-02	Daily	FRED DEXINUS (rupees per US dollar)
`stock_market_data`	4,896	2007-01-02 to 2026-10-06	Daily	NSE NIFTY 50 close (price return); BSE Sensex close
`inflation_data`	235	2007-01-01 to 2026-07-01	Monthly	IMF India CPI, base 2010=100, and year-on-year % change
`wholesale_price_data`	169	2012-04-01 to 2026-04-01	Monthly	Office of the Economic Adviser (DPIIT) WPI All Commodities, base 2011-12=100; inflation rate calculated from the index
`monetary_policy_data`	58	2006-10-30 to 2025-12-05	On change	RBI repo rate changes, compiled from public timelines
`fuel_price_data`	147	2006-11-30 to 2026-05-25	On change	PPAC Delhi retail price of petrol and diesel, one row per revision
`Total`	**28,075**			

Columns and meaning

Table	Columns	Meaning
`crude_oil_data`	`date, brent_oil_usd`	Daily Brent crude spot price in US dollars per barrel
`currency_data`	`date, usd_inr`	Rupees needed for 1 US dollar; a higher value means a weaker rupee
`stock_market_data`	`date, nifty50, sensex`	Daily closing values of India's two main stock indices
`inflation_data`	`month, cpi_index, inflation_rate`	Consumer price index and its year-on-year % change (month = first day of the month)
`wholesale_price_data`	`month, wpi_index, wpi_inflation_rate`	Wholesale price index and its year-on-year % change
`monetary_policy_data`	`effective_date, repo_rate`	RBI repo rate; one row for each date the rate changed
`fuel_price_data`	`date, petrol_price, diesel_price`	Delhi retail price in rupees per litre; one row per price revision

Table definitions (DDL)

The statements below create the schema's seven tables. Dates are stored as `DATE`, prices and indices as `DECIMAL`, so values such as 96.82 rupees per dollar are held exactly.

STEP 1 — RESET AND CREATE THE CRUDE OIL AND CURRENCY TABLES

```
DROP TABLE IF EXISTS crude_oil_data, currency_data, inflation_data,
    wholesale_price_data, monetary_policy_data, stock_market_data, fuel_price_data CASCADE;

CREATE TABLE crude_oil_data (
    date           DATE,
    brent_oil_usd  DECIMAL(10,2)
);

CREATE TABLE currency_data (
    date     DATE,
    usd_inr  DECIMAL(10,2)
);
```

STEP 2 — INFLATION AND WHOLESALE PRICE TABLES

```
CREATE TABLE inflation_data (
    month           DATE,
    cpi_index       DECIMAL(10,2),
    inflation_rate  DECIMAL(6,2)
);

CREATE TABLE wholesale_price_data (
    month               DATE,
    wpi_index           DECIMAL(10,2),
    wpi_inflation_rate  DECIMAL(6,2)
);
```

STEP 3 — POLICY, MARKET AND FUEL TABLES

```
CREATE TABLE monetary_policy_data (
    effective_date  DATE,
    repo_rate       DECIMAL(5,2)
);

CREATE TABLE stock_market_data (
    date     DATE,
    nifty50  DECIMAL(12,2),
    sensex   DECIMAL(12,2)
);

CREATE TABLE fuel_price_data (
    date          DATE,
    petrol_price  DECIMAL(10,2),
    diesel_price  DECIMAL(10,2)
);
```

How the tables connect

> **JOIN RULE**
>
> The tables have no foreign keys. They are joined on dates: daily tables on `date`, monthly tables on the first day of the month (`DATE_TRUNC('MONTH', date)`), and the two event tables with an "as of" or "next event" lookup. Never join a daily date directly to a monthly column.

SECTION 03

Data Quality and Limitations

What the data can and cannot support. These limits apply to every result in the document.

Coverage and overlap

- **Overlap window.** Brent, USD/INR and NIFTY 50 are all present together on 4,635 days (2007-01-02 to 2026-09-29), so the project has about 4,600 truly overlapping daily rows, not 10,000+. Brent and USD/INR alone overlap on 8,863 days.
- **Different end dates.** Brent ends 2026-09-29, USD/INR 2026-10-02, NIFTY 2026-10-05, Sensex 2026-10-06, CPI 2026-07, WPI 2026-04, repo changes 2025-12 and fuel prices 2026-05.
- **Partial year.** 2026 is a partial year, so its averages cover fewer months than other years.

Source and measurement

- **Brent is a spot price** (EIA), not a futures price, so it can differ from the prices quoted in news.
- **CPI and WPI have different base years** (2010=100 and 2011-12=100), so only their inflation rates are compared. WPI covers 2012-04 to 2026-04; the newer 2022-23 base series is not included. The first 12 WPI inflation rates are empty because no prior-year value exists.
- **CPI source.** CPI comes from the IMF (a linked series) and may differ slightly from the official MoSPI series.
- **Missing market values.** Sensex is empty on 42 days where NIFTY traded, and NIFTY is empty on 1 day where Sensex traded (a real trading-day mismatch, not an error).

Gaps and small samples

- **Fuel prices cover Delhi only,** for 2006-11 to 2017-06 and 2022-04 to 2026-05. The period 2017-06-02 to 2022-04-28 is not loaded; a row dated 2017-06-02 with empty prices marks the gap so lookups return empty instead of a stale price.
- **Repo rate history** is compiled from public timelines. A few 2011–2013 dates may be off by one day and should be checked against the RBI archive before publication.
- **Small samples.** Only 58 repo-rate rows and 7 fuel revisions in 2022–2026 exist, so results built on them are suggestive, not definitive.

> **ASSOCIATION IS NOT CAUSATION**
>
> Every finding describes what moved together or followed what, not what caused what.

SECTION 04

Exploration Queries

Short starter queries run on each table before the main analysis, to learn the shape of the data and to confirm the load.

Question asked of the data	Query (short form)	Result
Most expensive and cheapest oil	`SELECT MAX(brent_oil_usd), MIN(brent_oil_usd) FROM crude_oil_data;`	Highest $143.95 on 2008-07-03; lowest $9.10 on 1998-12-10
Day the rupee was weakest	`SELECT date, usd_inr FROM currency_data ORDER BY usd_inr DESC LIMIT 1;`	96.82 per USD on 2026-05-20
Days with no Sensex value	`SELECT COUNT(*) FROM stock_market_data WHERE sensex IS NULL;`	42 days
Five highest-inflation months	`SELECT month, inflation_rate FROM inflation_data ORDER BY inflation_rate DESC LIMIT 5;`	2010-01 (16.22%); 2009-12 (14.97%); 2010-02 (14.86%); 2010-03 (14.86%); 2010-05 (13.91%)
Most recent 3 WPI months	`SELECT * FROM wholesale_price_data ORDER BY month DESC LIMIT 3;`	2026-04: index 167.00, 8.30%; 2026-03: index 160.80, 3.88%; 2026-02: index 158.40, 2.26%
Months with no WPI inflation rate	`SELECT COUNT(*) FROM wholesale_price_data WHERE wpi_inflation_rate IS NULL;`	12 months
Highest and lowest repo rate	`SELECT MAX(repo_rate), MIN(repo_rate) FROM monetary_policy_data;`	9.00% and 4.00%
Latest fuel prices	`SELECT * FROM fuel_price_data ORDER BY date DESC LIMIT 3;`	2026-05-25: petrol 102.12, diesel 95.20; 2026-05-23: petrol 99.51, diesel 92.49; 2026-05-19: petrol 98.64, diesel 91.58
Fuel gap marker	`SELECT * FROM fuel_price_data WHERE petrol_price IS NULL;`	1 row (2017-06-02)

READING THE OUTPUT

The extremes confirm the load: the oil peak of $143.95 in July 2008, the 2026 rupee low of 96.82 per dollar, and the single fuel-gap marker row that Q10 relies on.

ANALYSIS QUESTION 01

Q1. Year-by-year scorecard

Business question. How did oil, the rupee and the stock market change each year, and in which years did all three move sharply?

Technique. `EXTRACT(YEAR)` , one CTE per table, `LAG` for year-on-year % change, `CASE` for the sharp-year label.

Approach

- Average each daily series by year from 2007 (the year NIFTY starts), so all three overlap.
- Use `LAG` to bring in the previous year and calculate the percent change: (this year − last year) / last year × 100.
- A year is called sharp when Brent moves more than 10%, USD/INR more than 3% and NIFTY more than 10%, in either direction. These limits are a choice and are stated with the result.

Q1 — YEARLY SCORECARD

```
WITH BRENT AS (
    SELECT EXTRACT(YEAR FROM date) AS year,
           ROUND(AVG(brent_oil_usd),2) AS avg_oil_price
    FROM crude_oil_data
    WHERE EXTRACT(YEAR FROM date) >= 2007
    GROUP BY EXTRACT(YEAR FROM date)
),
USD AS (
    SELECT EXTRACT(YEAR FROM date) AS year,
           ROUND(AVG(usd_inr),2) AS usd_average
    FROM currency_data
    WHERE EXTRACT(YEAR FROM date) >= 2007
    GROUP BY EXTRACT(YEAR FROM date)
),
STOCK AS (
    SELECT EXTRACT(YEAR FROM date) AS year,
           ROUND(AVG(nifty50),2) AS avg_of_nifty,
           ROUND(AVG(sensex),2)  AS avg_of_sensex
    FROM stock_market_data
    WHERE EXTRACT(YEAR FROM date) >= 2007
    GROUP BY EXTRACT(YEAR FROM date)
),
SCORECARD AS (
    SELECT b.year, b.avg_oil_price, u.usd_average, s.avg_of_nifty, s.avg_of_sensex,
           ROUND((b.avg_oil_price - LAG(b.avg_oil_price) OVER (ORDER BY b.year))
                 / LAG(b.avg_oil_price) OVER (ORDER BY b.year) * 100, 2) AS brent_change_pct,
           ROUND((u.usd_average - LAG(u.usd_average) OVER (ORDER BY b.year))
                 / LAG(u.usd_average) OVER (ORDER BY b.year) * 100, 2)   AS usd_change_pct,
           ROUND((s.avg_of_nifty - LAG(s.avg_of_nifty) OVER (ORDER BY b.year))
                 / LAG(s.avg_of_nifty) OVER (ORDER BY b.year) * 100, 2)  AS nifty_change_pct,
           ROUND((s.avg_of_sensex - LAG(s.avg_of_sensex) OVER (ORDER BY b.year))
                 / LAG(s.avg_of_sensex) OVER (ORDER BY b.year) * 100, 2) AS sensex_change_pct
    FROM BRENT b
    JOIN USD u   ON b.year = u.year
    JOIN STOCK s ON b.year = s.year
)
SELECT *,
       CASE
           WHEN (brent_change_pct > 10 OR brent_change_pct < -10)
            AND (usd_change_pct > 3    OR usd_change_pct < -3)
            AND (nifty_change_pct > 10 OR nifty_change_pct < -10)
           THEN 'ALL THREE SHARP'
           ELSE 'NOT ALL'
       END AS sharp_year
FROM SCORECARD
ORDER BY year;
```

Result

Year	Brent $	USD/INR	NIFTY	Sensex	Brent %	USD %	NIFTY %	Sensex %	Sharp?
2007	72.61	41.18	4,571	15,550	-	-	-	-	
2008	96.94	43.39	4,339	14,491	+33.51	+5.37	-5.08	-6.81	
2009	61.74	48.33	4,114	13,702	-36.31	+11.39	-5.19	-5.45	
2010	79.61	45.65	5,461	18,205	+28.94	-5.55	+32.75	+32.87	ALL THREE
2011	111.26	46.58	5,336	17,780	+39.76	+2.04	-2.29	-2.34	
2012	111.57	53.37	5,344	17,626	+0.28	+14.58	+0.15	-0.87	
2013	108.56	58.51	5,916	19,715	-2.70	+9.63	+10.71	+11.86	
2014	99.02	61.00	7,360	24,619	-8.79	+4.26	+24.42	+24.88	
2015	52.32	64.11	8,286	27,349	-47.16	+5.10	+12.58	+11.09	ALL THREE
2016	43.64	67.16	8,092	26,367	-16.59	+4.76	-2.34	-3.59	
2017	54.12	65.07	9,591	30,929	+24.01	-3.11	+18.52	+17.30	ALL THREE
2018	71.34	68.37	10,749	35,400	+31.82	+5.07	+12.08	+14.46	ALL THREE
2019	64.30	70.38	11,433	38,372	-9.87	+2.94	+6.36	+8.40	
2020	41.96	74.14	11,161	37,927	-34.74	+5.34	-2.37	-1.16	
2021	70.86	73.94	16,027	53,814	+68.88	-0.27	+43.59	+41.89	
2022	100.93	78.58	17,245	57,842	+42.44	+6.28	+7.60	+7.49	
2023	82.49	82.57	18,903	63,641	-18.27	+5.08	+9.61	+10.02	
2024	80.52	83.66	23,478	77,297	-2.39	+1.32	+24.20	+21.46	
2025	69.14	87.15	24,576	80,664	-14.13	+4.17	+4.68	+4.36	
2026	93.03	93.99	24,218	77,776	+34.55	+7.85	-1.45	-3.58	

What the data shows

Under this rule 4 years qualify: 2010, 2015, 2017, 2018. Over the whole window the rupee moved steadily from about 41 per dollar in 2007 to about 94 in 2026, and the NIFTY 50 average rose from about 4,600 to about 24,200. Brent was the most volatile of the three, for example +34.0% in 2008, −47.2% in 2015 and +68.9% in 2021. The five years named in the question (2008, 2013, 2020, 2022, 2026) do not meet this rule, because in each at least one of the three averages moved less than its threshold.

> **CAUTION**
>
> Yearly averages smooth out large swings inside a year, and 2026 is a partial year. The result depends on the thresholds chosen.

ANALYSIS QUESTION 02

Q2. Oil versus rupee relationship

Business question. How strongly do monthly Brent changes and monthly USD/INR changes move together, and is the relationship stable over time?

Technique. Monthly averages, % change with `LAG`, `CORR()` for the full period and over rolling 36-month windows.

> **HOW TO READ A CORRELATION**
>
> `CORR` gives a number from −1 to +1. Close to +1 means the two series move together; close to 0 means almost no link; close to −1 means they move in opposite directions. Because a higher USD/INR means a weaker rupee, a positive value would mean oil up with rupee weaker.

Q2 — FULL-PERIOD AND ROLLING CORRELATION

```
WITH BRENT_MONTHLY AS (
    SELECT DATE_TRUNC('MONTH', date)::DATE AS month,
           ROUND(AVG(brent_oil_usd),2) AS avg_oil_price
    FROM crude_oil_data
    GROUP BY DATE_TRUNC('MONTH', date)
),
USD_MONTHLY AS (
    SELECT DATE_TRUNC('MONTH', date)::DATE AS month,
           ROUND(AVG(usd_inr),2) AS avg_usd_price
    FROM currency_data
    GROUP BY DATE_TRUNC('MONTH', date)
),
MONTHLY AS (
    SELECT b.month, b.avg_oil_price, u.avg_usd_price,
           ROUND((b.avg_oil_price - LAG(b.avg_oil_price) OVER (ORDER BY b.month))
                 / LAG(b.avg_oil_price) OVER (ORDER BY b.month) * 100, 2) AS oil_price_change_by_month,
           ROUND((u.avg_usd_price - LAG(u.avg_usd_price) OVER (ORDER BY b.month))
                 / LAG(u.avg_usd_price) OVER (ORDER BY b.month) * 100, 2) AS usd_changes_by_month
    FROM BRENT_MONTHLY b
    JOIN USD_MONTHLY u ON b.month = u.month
)
-- (a) one number for the full period
SELECT ROUND(CORR(oil_price_change_by_month, usd_changes_by_month)::NUMERIC, 4) AS correlation
FROM MONTHLY;

-- (b) rolling 36-month correlation, one value per month
SELECT month, oil_price_change_by_month, usd_changes_by_month,
       ROUND(CORR(oil_price_change_by_month, usd_changes_by_month)
             OVER (ORDER BY month ROWS BETWEEN 35 PRECEDING AND CURRENT ROW)::NUMERIC, 4)
             AS rolling_36m_correlation
FROM MONTHLY
ORDER BY month;
```

Result

Measure	Value
Months in the sample	473 (1987-05 to 2026-09)
Correlation, full period	about −0.12 (very weak, slightly negative)
Rolling 36-month, first full window	1990-05: −0.08
Rolling 36-month, lowest	2012-09: −0.62
Rolling 36-month, highest	2026-04: 0.54
Rolling 36-month, latest	2026-09: 0.37
Share of rolling windows above zero	37.8% of 437 windows

What the data shows

Over the full period, monthly oil changes and monthly rupee changes show almost no relationship (about −0.12). The rolling value is not constant: it swings from clearly negative (about −0.62 in 2012) to clearly positive (about +0.54 in 2026). The link is therefore weak overall and not stable over time.

> **CAUTION**
>
> Before the early 1990s the rupee rate was largely administered, so the oldest windows behave differently from the market-driven years.

ANALYSIS QUESTION 03

Q3. What follows an oil spike?

Business question. In months when Brent rose more than 10%, what happened to the rupee and NIFTY in the same month and the next 1 to 3 months?

Technique. Monthly aggregation, `LAG` for monthly % change, `LEAD` to look 1 to 3 months ahead, `CASE` to label spike months, conditional `SUM` and `AVG`.

Approach

- A spike month is a month where Brent's monthly average rose by more than 10% over the previous month.
- `LEAD` is calculated before filtering. Filtering first would make `LEAD` jump to the next spike instead of the next month.
- For each window the query counts how many spike months had a rupee fall (USD/INR up) or a NIFTY rise, and the average change.

Q3 — SPIKE MONTHS AND WHAT FOLLOWS

```
-- BRENT, USD and STOCK blocks: monthly averages (same pattern as Q2), then:
MONTHLY_DATA AS (
    SELECT b.month, b.avg_oil_price, u.avg_usd_inr, s.nifty50_avg,
           ROUND((b.avg_oil_price - LAG(b.avg_oil_price) OVER (ORDER BY b.month))
                 / LAG(b.avg_oil_price) OVER (ORDER BY b.month) * 100, 2) AS oil_change_percent,
           ROUND((u.avg_usd_inr - LAG(u.avg_usd_inr) OVER (ORDER BY b.month))
                 / LAG(u.avg_usd_inr) OVER (ORDER BY b.month) * 100, 2)   AS usd_change_percent,
           ROUND((s.nifty50_avg - LAG(s.nifty50_avg) OVER (ORDER BY b.month))
                 / LAG(s.nifty50_avg) OVER (ORDER BY b.month) * 100, 2)   AS nifty50_change_percent
    FROM BRENT b
    JOIN USD u   ON b.month = u.month
    JOIN STOCK s ON b.month = s.month
),
OIL_ANALYSIS AS (
    SELECT *, CASE WHEN oil_change_percent > 10 THEN 'OIL SPIKE'
                   ELSE 'NO OIL SPIKE' END AS oil_spike_status
    FROM MONTHLY_DATA
),
FUTURE_ANALYSIS AS (
    SELECT *,
           LEAD(usd_change_percent, 1)  OVER (ORDER BY month) AS usd_change_next_month,
           LEAD(usd_change_percent, 2)  OVER (ORDER BY month) AS usd_change_after_2_month,
           LEAD(usd_change_percent, 3)  OVER (ORDER BY month) AS usd_change_after_3_month,
           LEAD(nifty50_change_percent, 1) OVER (ORDER BY month) AS nifty_change_next_month,
           LEAD(nifty50_change_percent, 2) OVER (ORDER BY month) AS nifty_change_after_2_month,
           LEAD(nifty50_change_percent, 3) OVER (ORDER BY month) AS nifty_change_after_3_month
    FROM OIL_ANALYSIS
)
SELECT
    ROUND(AVG(usd_change_percent),2) AS usd_avg_same_month,
    SUM(CASE WHEN usd_change_percent > 0 THEN 1 ELSE 0 END) AS usd_up_same_month,
    ROUND(AVG(usd_change_next_month),2) AS usd_avg_next_month,
    SUM(CASE WHEN usd_change_next_month > 0 THEN 1 ELSE 0 END) AS usd_up_next_month,
    -- ... the same pair of lines for +2 and +3 months, and for NIFTY ...
    ROUND(AVG(nifty_change_after_3_month),2) AS nifty_avg_after_3_month,
    SUM(CASE WHEN nifty_change_after_3_month > 0 THEN 1 ELSE 0 END) AS nifty_up_after_3_month
FROM FUTURE_ANALYSIS
WHERE oil_spike_status = 'OIL SPIKE';
```

Result (23 spike months)

Window after spike	Spike months with data	USD/INR rose (rupee weaker)	Avg USD/INR change %	NIFTY rose	Avg NIFTY change %
Same month	23	12	+0.12	18	+3.06
+1 month	22	14	+0.12	14	+2.14
+2 month	22	10	+0.16	10	+0.21
+3 month	22	11	+0.13	13	+0.88
All months (baseline)	237	134 (56.5%)	+0.34	145 of 236 (61.4%)	+0.87

What the data shows

After the 23 spike months the rupee did not weaken more often than in a normal month (it rose in 12 of 23 spike months versus 56.5% of all months), and the average rupee change after a spike was smaller than the all-month average. NIFTY rose in 18 of 23 spike months (78%, against 61% normally) but was weakest two months later (10 of 22 up, average +0.21%). The follow-through is inconsistent: it changes from month to month.

> **CAUTION**
>
> Only 23 events. The last spike (2026-09) has no future months yet, so the +1, +2 and +3 columns are out of 22.

ANALYSIS QUESTION 04

Q4. The worst rupee days

Business question. On the 20 biggest one-day rupee falls, what was Brent doing in the days around them, and what did NIFTY do?

Technique. `LAG` for daily % change, `ORDER BY` with `LIMIT` for the top 20, LEFT JOINs to Brent and NIFTY daily changes.

Approach

- A rupee fall means USD/INR rises, so the worst days are the largest positive daily changes since 2007-01-02 (when NIFTY data begins).
- `LAG` is calculated on each table's full history first, so each day is compared with its own previous trading day.
- LEFT JOINs keep all 20 days even if one market has no data that day.

Q4 — TOP 20 ONE-DAY RUPEE FALLS

```
WITH RUPEE_DAILY_CHANGE AS (
    SELECT date, usd_inr,
           LAG(usd_inr) OVER (ORDER BY date) AS previous_usd_inr,
           ROUND((usd_inr - LAG(usd_inr) OVER (ORDER BY date))
                 / LAG(usd_inr) OVER (ORDER BY date) * 100, 2) AS rupee_change_percent
    FROM currency_data
),
WORST_DAYS AS (
    SELECT date, usd_inr, previous_usd_inr, rupee_change_percent
    FROM RUPEE_DAILY_CHANGE
    WHERE date >= '2007-01-02'
    ORDER BY rupee_change_percent DESC
    LIMIT 20
),
BRENT_DAILY AS (
    SELECT date, brent_oil_usd,
           ROUND((brent_oil_usd - LAG(brent_oil_usd,1) OVER (ORDER BY date))
                 / LAG(brent_oil_usd,1) OVER (ORDER BY date) * 100, 2) AS brent_1day_change_percent,
           ROUND((brent_oil_usd - LAG(brent_oil_usd,5) OVER (ORDER BY date))
                 / LAG(brent_oil_usd,5) OVER (ORDER BY date) * 100, 2) AS brent_5day_change_percent
    FROM crude_oil_data
),
NIFTY_DAILY AS (
    SELECT date, nifty50,
           ROUND((nifty50 - LAG(nifty50,1) OVER (ORDER BY date))
                 / LAG(nifty50,1) OVER (ORDER BY date) * 100, 2) AS nifty_1day_change_percent
    FROM stock_market_data
)
SELECT w.date, b.brent_oil_usd, b.brent_1day_change_percent, b.brent_5day_change_percent,
       n.nifty50, n.nifty_1day_change_percent,
       w.usd_inr, w.previous_usd_inr, w.rupee_change_percent
FROM WORST_DAYS w
LEFT JOIN BRENT_DAILY b ON w.date = b.date
LEFT JOIN NIFTY_DAILY n ON w.date = n.date
ORDER BY w.rupee_change_percent DESC;
```

Result (20 days)

Date	USD/INR	Rupee %	Brent $	Brent 1-day %	Brent 5-day %	NIFTY 1-day %
2008-11-12	49.20	+4.02	52.47	-4.18	-14.11	-3.07
2013-08-28	68.80	+3.86	116.27	+0.92	+4.92	-0.05
2008-07-28	42.52	+3.45	125.67	+0.78	-2.84	+0.47
2013-08-27	66.24	+3.08	115.21	+2.66	+4.04	-3.45
2013-09-03	67.71	+3.04	115.49	-0.41	+2.90	-3.77
2013-06-10	58.13	+2.76	103.87	-0.19	+2.20	-0.05
2008-09-02	44.40	+2.66	104.94	-7.53	-4.37	+3.57
2013-07-30	60.80	+2.48	107.47	-0.58	-1.65	-1.31
2013-08-19	63.13	+2.40	111.41	-0.37	+1.95	-1.69
2011-09-22	49.47	+2.38	109.21	-4.42	-6.43	-4.08
2018-12-10	72.44	+2.24	60.29	-2.30	+0.20	-1.92
2009-02-17	49.60	+2.10	39.69	-8.46	-15.96	-2.74
2013-11-12	63.73	+2.00	106.29	+0.50	+1.22	-1.00
2008-12-23	48.66	+1.99	35.27	-7.38	-16.02	-2.32
2008-09-16	46.81	+1.92	85.85	-5.09	-13.23	+0.05
2012-10-09	52.73	+1.85	114.32	+1.53	+1.63	+0.50
2009-03-02	51.80	+1.81	42.60	-4.08	+3.22	-3.22
2010-08-23	46.50	+1.75	73.08	-0.54	-1.98	+0.23
2013-06-26	60.70	+1.74	100.62	-0.88	-4.68	-0.36
2007-10-10	39.13	+1.69	77.85	+0.04	+0.06	+2.14

What the data shows

Brent rose on the day on only 6 of 20 days and was up over the previous five trading days on 10 of 20, which is about a coin flip. NIFTY fell on 14 of 20 days. The dates cluster in two years: 2013 (8 days) and 2008 (5 days), 13 of the 20 days. In late 2008 Brent was falling sharply (for example −14.1% over five days on 2008-11-12) while the rupee had one of its worst days. Oil does not consistently appear around the worst rupee days, so other events appear to dominate.

> **CONTEXT AND CAUTION**
>
> These two years coincide with the 2008 global financial crisis and the 2013 emerging-market sell-off. That context comes from outside the dataset and should be verified before it is quoted; the data alone cannot say why the rupee fell.

ANALYSIS QUESTION 05

Q5. Oil-to-inflation delay

Business question. Does CPI or WPI inflation follow Brent with a lag of 0 to 6 months, and which lag fits best?

Technique. Monthly % changes, `LAG` shifts of the oil series, `CORR()` for each lag.

Approach

- Calculate the monthly % change of Brent, CPI and WPI, each inside its own block.
- Shift the oil change back by 0 to 6 months with `LAG`, so 'lag 2' compares oil two months ago with inflation today.
- Take the correlation for each lag and compare CPI with WPI on the same months (2012-04 to 2026-04).

Q5 — CORRELATION AT EACH LAG

```
-- Monthly % change for Brent, CPI and WPI (each computed in its own block, then joined)
-- BRENT_MONTHLY -> BRENT (oil_change_percent)
-- CPI_MONTHLY   -> CPI   (cpi_change_percent)
-- WPI_MONTHLY   -> WPI   (wpi_change_percent)
SHIFTED AS (
    SELECT b.month, b.oil_change_percent, c.cpi_change_percent, w.wpi_change_percent,
           LAG(b.oil_change_percent,1) OVER (ORDER BY b.month) AS oil_lag_1,
           LAG(b.oil_change_percent,2) OVER (ORDER BY b.month) AS oil_lag_2,
           LAG(b.oil_change_percent,3) OVER (ORDER BY b.month) AS oil_lag_3,
           LAG(b.oil_change_percent,4) OVER (ORDER BY b.month) AS oil_lag_4,
           LAG(b.oil_change_percent,5) OVER (ORDER BY b.month) AS oil_lag_5,
           LAG(b.oil_change_percent,6) OVER (ORDER BY b.month) AS oil_lag_6
    FROM BRENT b
    JOIN CPI c ON b.month = c.month
    JOIN WPI w ON w.month = c.month
)
SELECT
    ROUND(CORR(oil_change_percent, cpi_change_percent)::NUMERIC,4) AS cpi_corr_lag_0,
    ROUND(CORR(oil_lag_1, cpi_change_percent)::NUMERIC,4)          AS cpi_corr_lag_1,
    -- ... lags 2 to 5 follow the same pattern ...
    ROUND(CORR(oil_lag_6, cpi_change_percent)::NUMERIC,4)          AS cpi_corr_lag_6,
    ROUND(CORR(oil_change_percent, wpi_change_percent)::NUMERIC,4) AS wpi_corr_lag_0,
    ROUND(CORR(oil_lag_1, wpi_change_percent)::NUMERIC,4)          AS wpi_corr_lag_1,
    -- ... lags 2 to 5 follow the same pattern ...
    ROUND(CORR(oil_lag_6, wpi_change_percent)::NUMERIC,4)          AS wpi_corr_lag_6
FROM SHIFTED;
```

Result (169 months, 2012-04 to 2026-04)

Lag (months)	CPI correlation	WPI correlation
0	−0.024	0.412
1	0.053	**0.591**
2	**0.096**	0.260
3	0.064	0.080
4	0.042	−0.057
5	−0.067	−0.050
6	−0.147	0.020

What the data shows

WPI inflation followed Brent most closely with a 1-month delay (correlation about 0.59), a fairly clear link. CPI inflation showed only a very weak link, best at a 2-month delay (about 0.10). On the longer CPI-only window (2007-01 to 2026-07) the best lag was 3 months at about 0.16, still weak.

> CONTEXT
>
> WPI measures wholesale prices, which include fuel and other oil products directly, while CPI is dominated by food and services. This explanation is general background, not something tested in the data.

ANALYSIS QUESTION 06

Q6. Inflation and markets under different RBI stances

Business question. Label each month as hiking, cutting or holding. How do CPI inflation, WPI inflation, USD/INR and NIFTY behave in each stance?

Technique. `LAG` on the repo-rate table, `CASE` for the stance label, `DISTINCT` , LEFT JOIN with `COALESCE` , `GROUP BY` .

Approach

- A repo-rate change that raised the rate makes that month HIKING; one that lowered it makes it CUTTING. Months with no change are HOLDING.
- `DISTINCT` keeps one row for each month, because two months (2008-06 and 2010-07) had two hikes.
- LEFT JOIN to the 235 inflation months with `COALESCE(stance, 'HOLDING')` labels the holding months.

Q6 — STANCE LABELS AND MONTHLY AVERAGES

```
WITH RATE_CHANGES AS (
    SELECT repo_rate, effective_date AS start_date,
           LAG(repo_rate) OVER (ORDER BY effective_date) AS previous_rate
    FROM monetary_policy_data
),
LABELLED AS (
    SELECT repo_rate, previous_rate, start_date,
           CASE WHEN repo_rate > previous_rate THEN 'HIKING'
                WHEN repo_rate < previous_rate THEN 'CUTTING'
                ELSE 'START' END AS stance
    FROM RATE_CHANGES
),
STANCE_MONTHS AS (
    SELECT DISTINCT DATE_TRUNC('MONTH', start_date)::DATE AS month, stance
    FROM LABELLED
    WHERE stance <> 'START'
),
-- USD_MONTHLY / USD_CHANGE and NIFTY_MONTHLY / NIFTY_CHANGE: monthly % change (as in Q2, Q3)
MONTH_TABLE AS (
    SELECT i.month, i.inflation_rate, w.wpi_inflation_rate,
           u.usd_change_percent, n.nifty_change_percent,
           COALESCE(s.stance, 'HOLDING') AS stance
    FROM inflation_data i
    LEFT JOIN STANCE_MONTHS s         ON i.month = s.month
    LEFT JOIN wholesale_price_data w  ON i.month = w.month
    LEFT JOIN USD_CHANGE u            ON i.month = u.month
    LEFT JOIN NIFTY_CHANGE n          ON i.month = n.month
)
SELECT stance, COUNT(*) AS months,
       ROUND(AVG(inflation_rate),2)      AS avg_cpi_inflation,
       ROUND(AVG(wpi_inflation_rate),2)  AS avg_wpi_inflation,
       ROUND(AVG(usd_change_percent),2)  AS avg_usd_change,
       ROUND(AVG(nifty_change_percent),2) AS avg_nifty_change
FROM MONTH_TABLE
GROUP BY stance
ORDER BY stance;
```

Result (235 months)

Stance	Months	CPI inflation %	WPI inflation %	USD/INR change %	NIFTY change %
CUTTING	28	6.05	0.12	+0.77	−0.89
HIKING	27	8.48	8.47	+0.41	−0.03
HOLDING	180	6.27	3.06	+0.26	+1.30

What the data shows

Inflation was highest in hiking months (CPI about 8.5%, WPI about 8.5%) and lowest in cutting months for WPI (about 0.1%). NIFTY did best in holding months (+1.30% a month) and worst in cutting months (−0.89%), and the rupee weakened the most in cutting months (+0.77%).

> **CAUTION**
>
> RBI tends to raise rates because inflation is already high, so hikes and high inflation go together; the table does not show that hikes caused the pattern. Cutting months include stress periods such as 2008–09 and 2020, which may explain the weak NIFTY and rupee in that group.

ANALYSIS QUESTION 07

Q7. Markets after rate decisions

Business question. What was NIFTY's and Sensex's return 1, 3 and 6 months after rate hikes versus rate cuts?

Technique. Event study with monthly averages, `LEAD` for future values, JOIN to the event months, `GROUP BY` stance.

Approach

- Build a monthly table with each index's value now and 1, 3 and 6 months later (`LEAD`), before filtering to the event months.
- Return = (later value − value in the event month) / value in the event month × 100.
- Average the returns for the 27 hike months and the 28 cut months.

Q7 — FORWARD RETURNS AFTER HIKES AND CUTS

```
-- NIFTY_MONTHLY, SENSEX_MONTHLY: monthly averages; STANCE_MONTHS: as in Q6
MARKET_FUTURE AS (
    SELECT n.month, n.avg_nifty,
           LEAD(n.avg_nifty,1) OVER (ORDER BY n.month) AS nifty_after_1,
           LEAD(n.avg_nifty,3) OVER (ORDER BY n.month) AS nifty_after_3,
           LEAD(n.avg_nifty,6) OVER (ORDER BY n.month) AS nifty_after_6,
           x.avg_sensex,
           LEAD(x.avg_sensex,1) OVER (ORDER BY n.month) AS sensex_after_1,
           LEAD(x.avg_sensex,3) OVER (ORDER BY n.month) AS sensex_after_3,
           LEAD(x.avg_sensex,6) OVER (ORDER BY n.month) AS sensex_after_6
    FROM NIFTY_MONTHLY n
    JOIN SENSEX_MONTHLY x ON n.month = x.month
),
EVENT_RETURNS AS (
    SELECT s.month, s.stance,
           ROUND((f.nifty_after_1 - f.avg_nifty) / f.avg_nifty * 100, 2)   AS nifty_return_1m,
           ROUND((f.nifty_after_3 - f.avg_nifty) / f.avg_nifty * 100, 2)   AS nifty_return_3m,
           ROUND((f.nifty_after_6 - f.avg_nifty) / f.avg_nifty * 100, 2)   AS nifty_return_6m,
           ROUND((f.sensex_after_1 - f.avg_sensex) / f.avg_sensex * 100, 2) AS sensex_return_1m,
           ROUND((f.sensex_after_3 - f.avg_sensex) / f.avg_sensex * 100, 2) AS sensex_return_3m,
           ROUND((f.sensex_after_6 - f.avg_sensex) / f.avg_sensex * 100, 2) AS sensex_return_6m
    FROM STANCE_MONTHS s
    JOIN MARKET_FUTURE f ON s.month = f.month
)
SELECT stance, COUNT(*) AS events,
       ROUND(AVG(nifty_return_1m),2)  AS avg_nifty_1m,
       ROUND(AVG(nifty_return_3m),2)  AS avg_nifty_3m,
       ROUND(AVG(nifty_return_6m),2)  AS avg_nifty_6m,
       ROUND(AVG(sensex_return_1m),2) AS avg_sensex_1m,
       ROUND(AVG(sensex_return_3m),2) AS avg_sensex_3m,
       ROUND(AVG(sensex_return_6m),2) AS avg_sensex_6m
FROM EVENT_RETURNS
GROUP BY stance
ORDER BY stance;
```

Result

Stance	Events	NIFTY 1m %	NIFTY 3m %	NIFTY 6m %	Sensex 1m %	Sensex 3m %	Sensex 6m %
CUTTING	28	+1.22	+4.47	+12.23	+1.23	+4.65	+13.01
HIKING	27	−0.04	−0.11	+1.26	−0.09	−0.27	+1.06

What the data shows

After rate cuts, NIFTY and Sensex rose on average by about 1.2%, 4.5% and 12–13% over 1, 3 and 6 months. After rate hikes the average returns were close to zero for 1 and 3 months and about 1% after 6 months. The two indices tell the same story.

> **CAUTION**
>
> Only 27 hikes and 28 cuts, and cuts often came during or after market stress followed by recovery. The pattern is descriptive, not a forecast.

ANALYSIS QUESTION 08

Q8. High-oil versus normal-oil regimes

Business question. Group months into Brent buckets (under 60, 60 to 100, over 100 dollars). How do USD/INR change, NIFTY returns, CPI and WPI inflation compare across buckets?

Technique. `CASE` buckets, `GROUP BY` , `AVG` and `STDDEV` (STDDEV measures how much a value swings around its average).

Q8 — OIL PRICE BUCKETS

```
-- BRENT_MONTHLY, USD_CHANGE and NIFTY_CHANGE: monthly blocks as in Q2 and Q3
MONTH_TABLE AS (
    SELECT i.month, b.avg_brent,
           CASE WHEN b.avg_brent < 60   THEN '1. UNDER 60'
                WHEN b.avg_brent <= 100 THEN '2. 60 TO 100'
                ELSE '3. OVER 100' END AS oil_bucket,
           i.inflation_rate, w.wpi_inflation_rate,
           u.usd_change_percent, n.nifty_change_percent
    FROM inflation_data i
    JOIN BRENT_MONTHLY b            ON i.month = b.month
    LEFT JOIN wholesale_price_data w ON i.month = w.month
    LEFT JOIN USD_CHANGE u           ON i.month = u.month
    LEFT JOIN NIFTY_CHANGE n         ON i.month = n.month
)
SELECT oil_bucket, COUNT(*) AS months,
       ROUND(AVG(avg_brent),2)           AS avg_brent_price,
       ROUND(AVG(inflation_rate),2)      AS avg_cpi_inflation,
       ROUND(AVG(wpi_inflation_rate),2)  AS avg_wpi_inflation,
       ROUND(AVG(usd_change_percent),2)  AS avg_usd_change,
       ROUND(STDDEV(usd_change_percent),2) AS std_usd_change,
       ROUND(AVG(nifty_change_percent),2)  AS avg_nifty_change,
       ROUND(STDDEV(nifty_change_percent),2) AS std_nifty_change
FROM MONTH_TABLE
GROUP BY oil_bucket
ORDER BY oil_bucket;
```

Result (235 months)

Oil bucket	Months	Avg Brent $	CPI %	WPI %	USD chg %	USD swing	NIFTY chg %	NIFTY swing
1. UNDER 60	57	47.74	5.37	0.11	+0.09	1.41	+1.28	5.86
2. 60 TO 100	121	76.15	6.08	3.46	+0.23	1.52	+1.02	4.42
3. OVER 100	57	111.58	8.52	7.52	+0.82	2.15	+0.22	4.72

What the data shows

When oil was above $100 (57 months), CPI inflation averaged about 8.5% and WPI inflation about 7.5%, against about 5.4% and 0.1% when oil was under $60. The rupee weakened more on average in high-oil months (+0.82% a month) and swung more (2.15 against 1.41). NIFTY's average monthly change was smallest in high-oil months (+0.22%). The over-100 months fall in 2008 (6), 2011 (11), 2012 (11), 2013 (12), 2014 (8), 2022 (6), 2026 (3), which covers 2011–2014, 2022 and 2026.

> **CAUTION**
>
> High-oil periods also coincide with other conditions, so the comparison shows which things moved together, not what oil caused.

ANALYSIS QUESTION 09

Q9. Oil in rupee terms

Business question. Compute Brent in rupees (`brent_oil_usd` × `usd_inr`). For each year, how much of the change in the rupee cost of oil came from the dollar oil price and how much from a weaker rupee?

Technique. Yearly averages, % change, and an approximate decomposition: % change in rupee cost = % change in oil + % change in USD/INR.

> **WORKED EXAMPLE**
>
> If oil rises 10% and the dollar rises 5% against the rupee, the rupee cost of oil rises about 15.5% (10% + 5% + a 0.5% cross term).

Q9 — RUPEE COST OF OIL, DECOMPOSED

```
WITH DAILY_RUPEE_OIL AS (
    SELECT c.date, c.brent_oil_usd, u.usd_inr,
           c.brent_oil_usd * u.usd_inr AS brent_in_inr
    FROM crude_oil_data c
    JOIN currency_data u ON c.date = u.date
    WHERE c.date >= '2007-01-01'
),
YEARLY AS (
    SELECT EXTRACT(YEAR FROM date) AS year,
           ROUND(AVG(brent_oil_usd),2) AS avg_brent_usd,
           ROUND(AVG(usd_inr),2)       AS avg_usd_inr,
           ROUND(AVG(brent_in_inr),2)  AS avg_brent_inr
    FROM DAILY_RUPEE_OIL
    GROUP BY EXTRACT(YEAR FROM date)
),
CHANGES AS (
    SELECT year, avg_brent_inr,
           ROUND((avg_brent_usd - LAG(avg_brent_usd) OVER (ORDER BY year))
                 / LAG(avg_brent_usd) OVER (ORDER BY year) * 100, 2) AS oil_change_percent,
           ROUND((avg_usd_inr - LAG(avg_usd_inr) OVER (ORDER BY year))
                 / LAG(avg_usd_inr) OVER (ORDER BY year) * 100, 2)   AS usd_change_percent,
           ROUND((avg_brent_inr - LAG(avg_brent_inr) OVER (ORDER BY year))
                 / LAG(avg_brent_inr) OVER (ORDER BY year) * 100, 2) AS rupee_cost_change_percent
    FROM YEARLY
)
SELECT year, avg_brent_inr, oil_change_percent, usd_change_percent,
       rupee_cost_change_percent,
       ROUND(rupee_cost_change_percent - oil_change_percent - usd_change_percent, 2) AS cross_effect,
       CASE WHEN oil_change_percent IS NULL THEN 'NO PREVIOUS YEAR'
            WHEN ABS(usd_change_percent) > ABS(oil_change_percent) THEN 'CURRENCY'
            ELSE 'OIL PRICE' END AS main_driver
FROM CHANGES
ORDER BY year;
```

Result

Year	Brent in INR	Oil %	USD/INR %	Rupee cost %	Cross effect	Main driver
2007	2,966.29	-	-	-	-	NO PREVIOUS YEAR
2008	4,152.24	+34.01	+5.47	+39.98	+0.50	OIL PRICE
2009	2,965.07	−36.58	+11.36	−28.59	−3.37	OIL PRICE
2010	3,627.34	+29.02	−5.55	+22.34	−1.13	OIL PRICE
2011	5,177.67	+39.90	+2.04	+42.74	+0.80	OIL PRICE
2012	5,945.69	+0.28	+14.66	+14.83	−0.11	**CURRENCY**
2013	6,357.56	−2.70	+9.59	+6.93	+0.04	**CURRENCY**
2014	6,042.91	−8.60	+4.22	−4.95	−0.57	OIL PRICE
2015	3,347.55	−47.24	+5.10	−44.60	−2.46	OIL PRICE
2016	2,929.57	−16.66	+4.76	−12.49	−0.59	OIL PRICE
2017	3,518.77	+23.93	−3.11	+20.11	−0.71	OIL PRICE
2018	4,887.22	+31.96	+5.10	+38.89	+1.83	OIL PRICE
2019	4,528.36	−9.78	+2.91	−7.34	−0.47	OIL PRICE
2020	3,078.41	−35.17	+5.33	−32.02	−2.18	OIL PRICE
2021	5,237.51	+69.61	−0.26	+70.14	+0.79	OIL PRICE
2022	7,914.50	+42.51	+6.28	+51.11	+2.32	OIL PRICE
2023	6,811.96	−18.23	+5.09	−13.93	−0.79	OIL PRICE
2024	6,734.73	−2.36	+1.30	−1.13	−0.07	OIL PRICE
2025	6,017.89	−14.22	+4.20	−10.64	−0.62	OIL PRICE
2026	8,797.19	+35.25	+7.79	+46.18	+3.14	OIL PRICE

What the data shows

In 17 of the 19 years with a prior year, the change in the rupee cost of oil was driven more by the dollar oil price. In 2 years (2012 and 2013) the weaker rupee mattered more than oil: the dollar price of oil was almost flat while the rupee fell. These are the years of the rupee's sharpest slide in the yearly data.

> **CAUTION**
>
> The decomposition is approximate (the cross effect holds the remainder), and 2026 is a partial year.

ANALYSIS QUESTION 10

Q10. Fuel price pass-through

Business question. When Brent moved sharply, how quickly and by how much did Delhi petrol and diesel prices change? Compare 2006–2017 with 2022–2026.

Technique. Event windows, a correlated subquery for the next revision date, date subtraction, conditional `SUM` and `AVG`. The 2017–2022 gap is excluded.

Approach

- A big oil move is a month where Brent's monthly average changed by more than 10%, up or down.
- For each one, find the first fuel price revision after the last day of that month, and count the days in between.
- Count how many were followed by a revision within 30 days, and how many of those moved the same way as oil.

Q10 — DAYS FROM A BIG OIL MOVE TO THE NEXT REVISION

```
-- BRENT_MONTHLY / BRENT_CHANGE: monthly oil % change (as in Q2)
BIG_MOVES AS (
    SELECT month, oil_change_percent,
           (month + INTERVAL '1 MONTH' - INTERVAL '1 DAY')::DATE AS month_end,
           CASE WHEN month BETWEEN '2007-01-01' AND '2017-05-01' THEN '2007-2017'
                WHEN month >= '2022-05-01' THEN '2022-2026' END AS period
    FROM BRENT_CHANGE
    WHERE ABS(oil_change_percent) > 10
),
REVISIONS AS (
    SELECT date AS revision_date,
           petrol_price - LAG(petrol_price) OVER (ORDER BY date) AS petrol_change,
           diesel_price - LAG(diesel_price) OVER (ORDER BY date) AS diesel_change
    FROM fuel_price_data
),
REAL_REVISIONS AS (
    SELECT * FROM REVISIONS
    WHERE petrol_change IS NOT NULL AND diesel_change IS NOT NULL
),
NEXT_REV AS (
    SELECT b.month, b.period, b.oil_change_percent, b.month_end,
           (SELECT MIN(x.revision_date) FROM REAL_REVISIONS x
            WHERE x.revision_date > b.month_end) AS next_revision_date
    FROM BIG_MOVES b
    WHERE b.period IS NOT NULL
),
EVENTS AS (
    SELECT n.month, n.period, n.oil_change_percent, n.next_revision_date,
           (n.next_revision_date - n.month_end) AS delay_days, r.petrol_change
    FROM NEXT_REV n
    LEFT JOIN REAL_REVISIONS r ON r.revision_date = n.next_revision_date
)
SELECT period, COUNT(*) AS big_brent_moves,
       SUM(CASE WHEN delay_days <= 30 THEN 1 ELSE 0 END) AS revised_within_30_days,
       ROUND(AVG(delay_days),0)       AS avg_delay_days,
       ROUND(AVG(petrol_change),2)    AS avg_petrol_change,
       SUM(CASE WHEN delay_days <= 30
                 AND SIGN(oil_change_percent) = SIGN(petrol_change)
                THEN 1 ELSE 0 END)    AS same_direction_within_30
FROM EVENTS
GROUP BY period
ORDER BY period;
```

Result

Period	Big Brent moves	Revised within 30 days	Avg delay (days)	Avg petrol change (INR)	Same direction within 30 days
2007–2017	24	17	24	−0.02	14
2022–2026	9	1	315	+0.14	1

What the data shows

In 2007–2017, 17 of 24 big Brent moves were followed by a Delhi price revision within 30 days (average delay 24 days), and 14 of those moved in the same direction as oil. In 2022–2026 only 1 of 9 was followed by a revision within 30 days, with an average delay of 315 days. Fuel prices reacted much more often and faster in the earlier period.

> **CAUTION**
>
> Only 7 real price revisions exist in 2022–2026, so that row is a small sample. The last two big moves (2026-06 and 2026-09) have no later revision in the data yet. The delay is measured from the end of the move month, and the data covers Delhi only.

SECTION 05

Power BI Dashboard

A one-page dashboard that summarises the project for a non-technical reader: six KPI cards across the top and six charts below.

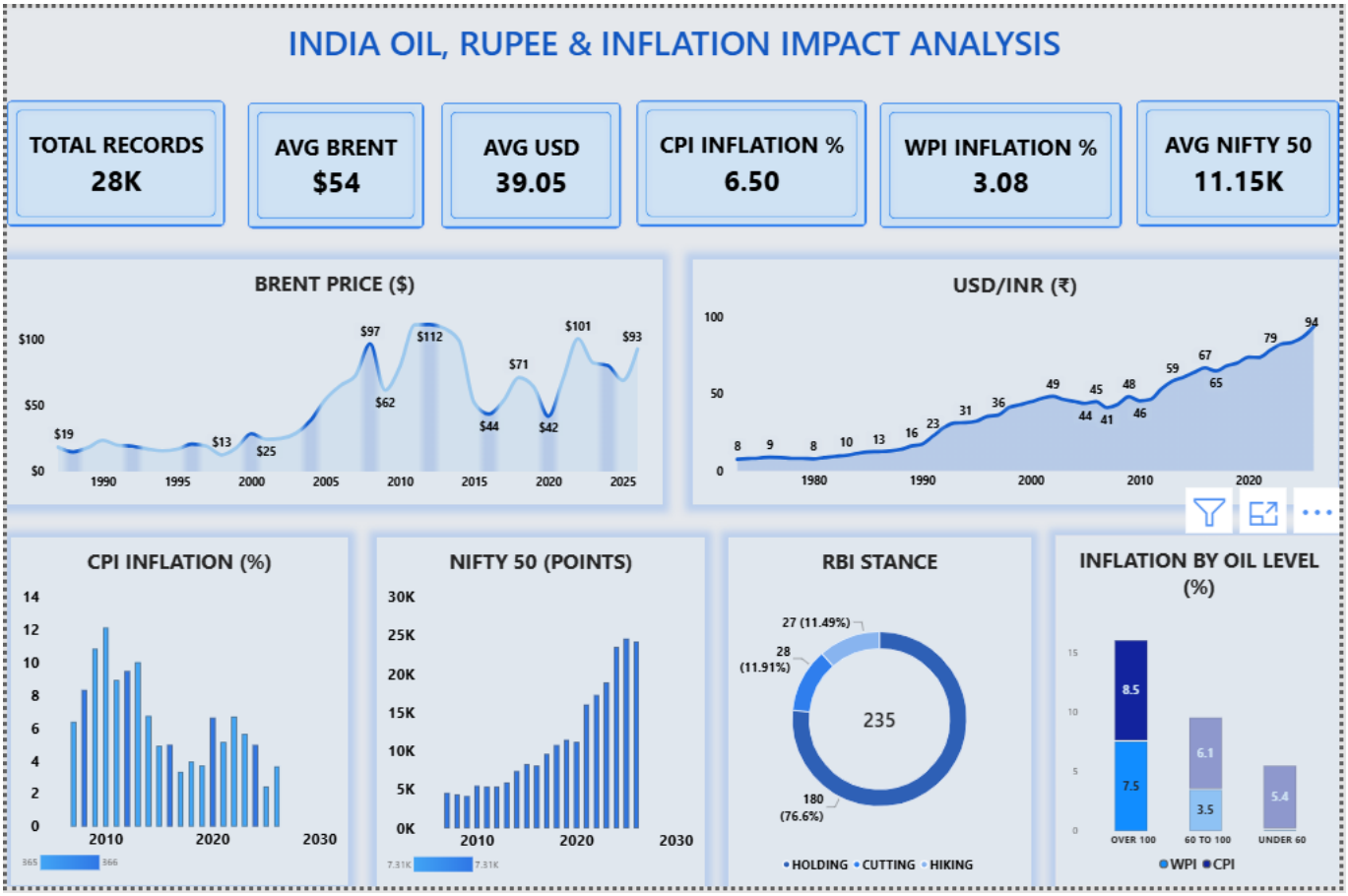


Figure 5.1 — One-page Power BI dashboard: six KPI cards across the top and six charts below.

Visual	Type	Built from
Total records, Avg Brent, Avg USD/INR, CPI inflation %, WPI inflation %, Avg NIFTY 50	KPI cards	The database tables (Average and row count)
Brent price by year	Line chart	crude_oil_data
USD/INR by year	Line chart	currency_data
CPI inflation by year	Column chart	inflation_data
NIFTY 50 by year	Column chart	stock_market_data
RBI stance (months)	Donut chart	Result of Q6 (28 cutting, 27 hiking, 180 holding)
Inflation by oil level	Clustered column chart	Result of Q8 (CPI and WPI by oil bucket)

The Power BI file reads the seven tables from the database export. The two charts built on SQL results (Q6 and Q8) use the result tables from the queries in this document.

SECTION 06

Key Findings, Run Guide and Roadmap

What the project found, how to rebuild it, and where it could go next.

Key findings

Question	Finding from the data
Q1 Scorecard	Rupee moved from about 41 to 94 per dollar and NIFTY from about 4,600 to 24,200 (2007–2026); only 2010, 2015, 2017 and 2018 met the all-three-sharp rule
Q2 Oil vs rupee	Monthly correlation about −0.12 overall; rolling 36-month value ranges from about −0.62 to +0.54, so the link is weak and unstable
Q3 After a spike	23 oil-spike months; the rupee did not weaken more often than normal and NIFTY follow-through was inconsistent
Q4 Worst rupee days	Brent rose on only 6 of 20; 13 of the 20 days fall in 2008 and 2013, so other events appear to dominate
Q5 Inflation delay	WPI best matches Brent at a 1-month lag (about 0.59); CPI link is weak (best about 0.10 at 2 months)
Q6 RBI stance	Inflation highest in hiking months (about 8.5%); NIFTY best in holding months, worst in cutting months
Q7 After rate decisions	After cuts NIFTY averaged about +12% over 6 months, after hikes about +1% (27 and 28 events)
Q8 Oil regimes	Over-$100 oil months had higher CPI and WPI inflation (8.5% and 7.5%) and a more volatile rupee
Q9 Rupee cost of oil	Oil price drove the rupee cost in most years; the weaker rupee dominated in 2012 and 2013
Q10 Fuel pass-through	17 of 24 big oil moves were followed by a price revision within 30 days in 2007–2017, against 1 of 9 in 2022–2026

How to run the project

Create an empty database, run the setup script in the Query Tool, then check the load with row counts. The setup script creates the 7 tables and loads 28,075 rows, and it can be re-run safely.

SETUP — PGADMIN AND VERIFICATION

```
-- 1. In pgAdmin create an empty database
CREATE DATABASE india_oil_rupee_inflation;

-- 2. Open the Query Tool on that database and run the setup script
--    (creates the 7 tables and loads 28,075 rows; it can be re-run safely)

-- 3. Check the load
SELECT COUNT(*) FROM crude_oil_data;        -- 9,097
SELECT COUNT(*) FROM currency_data;         -- 13,473
SELECT COUNT(*) FROM stock_market_data;     -- 4,896
SELECT COUNT(*) FROM inflation_data;        -- 235
SELECT COUNT(*) FROM wholesale_price_data;  -- 169
SELECT COUNT(*) FROM monetary_policy_data;  -- 58
SELECT COUNT(*) FROM fuel_price_data;       -- 147

-- 4. Run the queries Q1 to Q10 from this document, then open the Power BI file
```

Suggested repository layout

REPOSITORY LAYOUT

```
india-oil-rupee-inflation/
├── README.md                     # overview, sources, findings, limitations
├── sql/
│   ├── 01_schema_and_data.sql    # tables and loaded data
│   ├── 02_exploration.sql
│   └── q01_scorecard.sql ... q10_fuel_pass_through.sql
├── data/                         # the CSV / Excel exports
├── dashboard/
│   ├── india_oil_inflation.pbix
│   └── dashboard.png
└── docs/
    └── project-documentation.pdf  # this document
```

Roadmap

Enhancement	Detail
Complete the fuel series	Load the PPAC daily prices for 2017-06 to 2022-04 to remove the gap and extend Q10
Link the two WPI bases	Join the 2011-12 and 2022-23 series so WPI covers 2026 onward
Live database connection	Point Power BI directly at PostgreSQL so the dashboard refreshes from the tables
More explanatory data	Add trade deficit, global dollar index and GDP to separate oil effects from other drivers
Statistical depth	Add regression or Granger-style lag tests alongside the correlations

Skills demonstrated

Relational schema design, sourcing and cleaning real public data, SQL aggregation, joins and CTEs, window functions (`LAG` , `LEAD` , rolling `CORR`), conditional logic, event-study design, honest handling of data gaps and small samples, and Power BI dashboard design.